Sujin Joanna Yoo

Ph.D Student · Computer Science / Apparel Design

Providence, RD

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Education

Brown University — *Providence, RD*

Ph.D. in Computer Science — *Sept.2026 - May.2031 (Expected)*

- Advisors: Dr. Adriana Schulz, Dr. Daniel Ritchie

Cornell University — *Ithaca, NY*

M.A. Apparel Design with a Minor in Computer Science — *Aug.2024 - May.2026*

- Advisors: Dr. Fatma Baytar, Dr. Steve Marschner

University of Southern California (USC) — *Los Angeles, CA*

B.S. Computer Science and Business Administration — *Aug.2021 - May.2024*

- Combined Bachelor of Science degree program

Los Angeles Trade-Technical College — *Los Angeles, CA*

A.S. Mathematics — *Jan.2019 - May.2021*

- Earned concurrently with full-time employment

Fashion Institute of Design and Merchandising (FIDM) — *Los Angeles, CA*

A.A. Fashion Design — *Mar.2015 - Jun.2017*

Research Experience

Layer-Based Garment Modeling for Textile Form Weaving — *Inria Centre CRAFT Team*

Advised by Dr. Melina Skouras and Dr. Adrien Bousseau — *2026*

- Developed a predictive model for post-wash woven fabric behavior and a geometric model for simulating textile form weaving
- Work accepted to the ACM Symposium on Computational Fabrication (SCF) 2026

Interactive Size-Chart and Automated Pattern Adjustment System for Home Sewers — *Cornell University*

Advised by Dr. Fatma Baytar and Dr. Steve Marschner (Master's Thesis) — *2024-2026*

- Developed a 3D interactive size-chart and automated pattern adjustment tool integrated with CLO3D to help home sewers with non-standard body types
- Awarded Alan D. Mathios Research and Service Grant, Cornell University ($1,000)
- 1st Place – HCD Graduate Student Poster Contest, Cornell University ($250)

Posture Descriptors (PARCS): Improving Posture Analysis for Garment Fit — *Cornell University*

Cornell Fashion & Body Tech Lab — *2025*

- Assisted with 3D body scanning, pose estimation, and data preparation

Developing Inclusive Swim Cap Design Using 3D Body Scanning — *Cornell University*

Cornell Fashion & Body Tech Lab — *2024*

- Assisted with 3D body scanning and used Geomagic software to clean, process, and analyze head-shape data

Teaching Experience

2025 **DEA 1140:Principles of Design Computing**, Graduate Student Lecturer, Cornell University
2025 **FSAD 1450: Introduction to Fashion Design**, Teaching Assistant, Cornell University
2024 **FSAD 1450: Introduction to Fashion Design**, Teaching Assistant, Cornell University
2024 **Integration of Artificial Intelligence in the Fashion Industry**, Guest Lecturer, StyleCAD USA
2022-2024 **Software Programming Instructor**, iANT Education, Los Angeles, CA

Professional Experience

2024-ing **Creative Director**, Joanna Blüm Studio
2022-2024 **Freelance Apparel Technical Designer**, Beautiful Day Wedding
2023 **Freelance Apparel Technical Designer**, Brochu Walker
2021-2022 **Freelance Senior Fashion Designer**, Ginger Green
2021 **Senior Fashion Designer**, Mystree
2018-2021 **Associate Fashion Designer**, Papermoon Clothing
2017-2018 **Associate Fashion Designer**, Oh Yes Fashion
2016-2017 **Assistant Fashion Designer**, Aime
2015-2016 **Graphic/Apparel Designer**, Young 2 Dress

Publications

Journal Articles / Full Conference Papers

Yoo, S.; Voorwinden, M.; Trezeux, R.; McQuillan, H.; Bousseau, A.; Skouras, M. Modeling and Predicting Post-Wash Shrinkage in Woven Fabric. *Proceedings of the 11th ACM Symposium on Computational Fabrication*, 2026. Accepted.

Short Papers / Posters

Yoo, S.; Lopez, M.; Baytar, F. Hyperbolic Tessellation as a Method for Optimized Material Use in Garment Design. *International Textile and Apparel Association (ITAA) Annual Conference Proceedings*, 2026. Accepted.

Yoo, S.; Lopez, M.; Marschner, S.; Baytar, F. Designing an Interactive Size Chart to Improve Fit in DIY Garment Making. *International Conference on Human–Computer Interaction (HCII)*, 2026. https://doi.org/10.1007/978-3-032-30552-7_29

Baytar, F.; Yang, Y.; Maher, M.; Rivera, I.; **Yoo, S.**; Patel, S. Implementing 3D Body Scanning Methods for Inclusive Swim Cap Design and Sizing. *International Conference on Human–Computer Interaction (HCII)*, 2025. Springer LNCS 15792. https://doi.org/10.1007/978-3-031-93505-3_1

McDonald, C.; Navodhya, P. U.; Ruderman, G.; Carmicino, B.; Baytar, F.; Maher, M.; **Yoo, S.** Posture Descriptors (PARCS): What We Know and the Gaps. *Proceedings of 3DBODY.TECH*, 2025. https://doi.org/10.15221/25.37

Awards, Fellowships, & Grants

2025 **Recipient, Oris Glisson Fellowship**, International Textile & Apparel Association *$ 1,000*
2025 **1st Place, HCD Graduate Student Poster Contest**, Cornell University *$ 250*
2024 **Recipient, Alan D. Mathios Research and Service Grant**, Cornell University *$ 1,000*
2023 **Outstanding Team Project Award (Android App)**, University of Southern California

Relevant Coursework

Brown University:

CSCI1420 Machine Learning, **CSCI2952** Computational Design and Fabrication

Cornell University:

CS5620 Intro to Computer Graphics, **CS5643** Physical Based Animation for Computer Graphics, **CS5621** Computer Graphics Practicum, **MATH4520** Classical Geometries and Modern Applications, **FSAD6025** Design for Change, **FSAD3650** New Technologies for Fashion Design

Other Institutions:

- Completed full lower-division mathematics sequence (Calculus I–III, Linear Algebra, Differential Equations)

- Completed core undergraduate computer science courses (Data Structures, Algorithms, Discrete Mathematics, Artificial Intelligence, etc.)

Skills

Research areas: Computer graphics and apparel design, computational fabrication, physics-based simulation, interactive geometric modeling, human–computer interaction

Languages & tools: C++, Python, C#, CUDA, PyTorch, TaichiLang, OpenGL, OpenCV, Qt, Unity, Blender, CLO3D, Browzwear, Adobe Illustrator, Git

Certifications: AWS Certified Cloud Practitioner (Verification No. CT40GH217J44Q1KF); Oracle Certified Associate, Java SE 8 Programmer (Verification No. 20425079OCAJSE8)